

EndoLite-ViT: A Lightweight Vision Transformer for Explainable Automated Gastrointestinal Disease Classification

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Abstract—Gastroesophageal Reflux Disease (GERD) and colorectal polyps should be detected early to avoid the development of serious complications. Automated endoscopic image analysis facilitates their timely diagnosis, but most Vision Transformer (ViT) models are computationally expensive and will not be applicable in clinical environments in real-time. The present paper presents EndoLite-ViT, a lightweight ViT that is used to classify GERD, GERD Normal, Polyp, and Polyp Normal images according to four classes. EndoLite-ViT is a small patch embedding model with a shallow attention architecture, with parameters of only 0.18M, and 95% accuracy on the GastroEndoNet dataset, comparable to and occasionally larger ViT-based baselines. The consistency of the performance is confirmed by 5-fold cross-validation, and the analyses based on SHAPs and LIME demonstrate that the model has been consistent in focusing on clinically relevant areas. These findings indicate that EndoLite-ViT offers a good balance of precision, performance, and interpretability, which makes it applicable to use in endoscopy systems of limited resources. Future directions involve the validation and testing of more heterogeneous datasets in multi-institution validation.

Keywords— GERD, Colorectal Polyps, Endoscopic Images, ViT transformer model, Lightweight ViT, EndoLite-ViT

I. INTRODUCTION

GERD and colorectal polyps are common gastrointestinal (GI) diseases in the dominant burden of global healthcare systems. The symptoms of heartburn, regurgitation, and more may lead to severe complications like erosive esophagitis and adenocarcinoma due to GERD, which is a result of the reflux of gastric acid into the esophagus [1], [2]. Colorectal polyps and especially adenomatous forms are the forerunners of colorectal cancer, which has been one of the leading causes of death associated with cancer across the world [3]. Early and proper diagnosis of these diseases is hence important in alleviation of complications, malignant transformation prevention and better patient outcomes [4]. The necessity to have convenient diagnostic tools is critical, particularly in low-resource areas like Bangladesh, where high-tech imaging techniques and professional interpretation are usually constrained.

Conventional diagnostic methods, such as assessment of symptoms, pH-metry, multiple intraluminal impedance (pH-MII), esophagogastroduodenoscopy (EGD), and colonoscopy, have significant sensitivity and specificity as well as consistency limitations. An example is that pH-metry can only detect 34.1% of the cases of pediatric GERD, whereas pH-MII can detect 57.4% [1]. Likewise, a normal colonoscopy can overlook as many as 26% of minute (<5 mm) or sessile polyps [3]. Such constraints are also extended in the high-volume clinical setting, where small changes like non-erosive GERD or flat polyps could be disregarded [1], [3], [5]. As the amount of endoscopic images is growing swiftly, manual interpretation of images has become laborious, subjective, and error-ridden. In such a way, AI-based methods have become of great interest due to their ability to offer rapid, objective, and reproducible diagnostic assistance [8].

Several more recent developments in deep learning, specifically, the convolutional neural networks (CNNs), have shown encouraging results in GI image analysis, such as polyp segmentation [6], [7], GERD classification [5], [11], and general disease detection [8], [9]. Recently, Vision Transformers (ViTs) have also become an option, which relies on self-attention to learn global contextual relationships. However, the existing ViT architectures can be computationally intensive, and as such, they cannot be utilized in clinical practices in real-time and in a resource-constrained environment.

Nevertheless, these developments still have major obstacles to the use of AI-based systems to detect GERD and polyps. Some of the challenges are the inadequacy of large annotated datasets, the absence of sufficiently generalizable evaluation protocols, and uninterpretable models that clinicians will trust [6], [12]-[14]. Other studies in the past tend to focus on architecture-specific advances, or uses small-scale data, which restricts reproducibility and prevents clinical adoption [9]. These gaps bring out the necessity of a lightweight, generalizable, and interpretable diagnostic solution.

Due to the limitations of the clinical setting, especially in developing areas, there is an urgent necessity to develop

models that would be more efficient in computation, more diagnostic, and more interpretable. This leads to the primary research question of this paper: Does a lightweight confirm transformer-based model trained by an in-house method with high-endoscopic image classification performance offer transparent and clinically interpretable explanations?

In order to answer this question, this paper introduces EndoLite-ViT, a lightweight Vision Transformer that was trained on the GastroEndoNet dataset. The major contributions are:

- 1) EndoLite-ViT, a small Vision Transformer with 0.18M parameters, trained on four-class GERD and colorectal polyp classification, which has similar performance to larger pretrained ViT models.
- 2) A detailed 5-fold cross-validation of the results, and a comparison to various ViT baselines, showing that EndoLite-ViT is as accurate as much less expensive than ViT.
- 3) SHAP and LIME-based integrated interpretability, which would contribute to better transparency and allow model predictions to be adjusted to clinician expectations.

II. RELATED WORK

Deep learning and especially CNNs have greatly enhanced automated detection of gastrointestinal GI diseases, including GERD and colorectal polyps. The most commonly used tools of traditional diagnostic methods include pH-metry and pH-MII, which have a limitation, namely, as reported by Ristic et al., the detection rates of pH-MII and pH-metry in pediatric GERD are 57.4 and 34.1, respectively, without the help of AI [1]. CNN-based models have performed well in the detection of GERD: GERD-VGGNet by Wang et al. was reported to be 99.2% accurate at low-severity GERD and 100% at severe and normal GERD, surpassing the performance of physicians [5], and the GerdNet-RF, a hybrid transfer learning and random forest model by Yen et al., was reported to be 92.5% accurate on three-class GERD classification [11].

More complex architectures have been used in the case of colorectal polyps. Qadir et al. utilized recapacitated Mask R-CNN, achieving 80% precision and 72.6% recall by taking into account polyp shape variability [7]. Focus U-Net, a two-channel attention-gated CNN that was trained on hybrid focal loss, has an average Dice score of 0.941 in CVC-ClinicDB [3]. ResUNet++ by Jha et al. has conditional random fields and test-time augmentation to achieve robust segmentation [6]. Wan et al. proposed CRH-YOLO with 88.8 precision and 90.7 mAP on Kvasir-SEG, which is good at perceiving small polyps [12], and Sahafi et al. used YOLO-V8 to process wireless capsule endoscopy and performed well despite scarce data [13]. Ensemble and attention mechanisms are also studied in recent works. The MF-CNN by Islam et al. used several CNN streams to enhance classification across both GERD and polyp datasets [9]. Siddiqui et al. used deep ensemble learning using various pretrained CNNs, which had been steadily improving endoscopic classification [10]. EndoAIM, created by Xu et al., is an attention-based multi-task model that can perform simultaneous detection and segmentation, which is clinically relevant [15]. Hossain et al. presented a system, ColonFormer, based on transformers that can detect polyps on the colon better than CNNs, which can effectively capture long-range interdependence in endoscopic video images [16]. Li et al. demonstrated the federated

learning of polyp classification in colorectal polyp classification where the training of the model can be developed across institutions in a position to preserve the data privacy [17].

In this respect Selvaraj et al. (2025) and El Khalfaoui et al. (2024) stressed the hybrid CNN-ViT models to examine multiple classes in polyps and they proved that it is improved to combine the local and global features instead of employing only one [18, 20]. Masum et al. (2025) suggested an ensemble transformer with ViT and Swin Transformer to detect GERD and polyps with XAI explainability [19]. Elkarazle et al. (2023) also augmented MA-NET with Mix-ViT to ensure the detection of small and flat lesions, which were also improved [21]. SHENER and his team (2025) designed GISegNet (segmentation) in addition to the transformer-based hybrid models (GI anomaly classification), which are highly accurate but at the same time cost-effective in terms of computational cost [22]. These works highlight the opportunities of ViTs in GI imaging, but they are not widely used, and specific lightweight ViTs are not available.

In spite of these developments, there are still a number of issues. The majority of them are disease-specific or two-class studies, which do not allow generalization [8]. Imbalance of data and absence of cross-dataset validation led to an inability to be robust, whereas the low interpretability limits clinical adoption [14]. In addition, transformer-based models are also effective but computationally intensive, and this limits their application in resource-limited environments. Hence, there exists a pressing need for lightweight ViT architectures with high accuracy, efficiency, and explainability on multiclass GI disease classification to bridge the gaps in the performance as well as in the practical implementation.

TABLE I. KEY RESEARCH GAPS IN RECENT GI DISEASE DETECTION STUDIES

Study, Method (Year)	Limitations Identified	Resulting Research Gap
Nie et al., DL Review (2024) [8]	Reports major issues such as dataset imbalance, low generalization, and limited interpretability across GI imaging studies	Need for models that improve robustness and provide transparent, clinically meaningful explanations
Islam et al., MF-CNN (2024) [9]	Multi-stream CNN is accurate but more complex in architecture and not interpretable	Simpler, unified, and interpretable GI classification models are required
Wan et al., CRH-YOLO (2024) [12]	Strong for small polyp detection and only on single-task, not multiclass GI disease classification ones	Need end-to-end multiclass diagnostic systems covering both GERD and polyps
Xu et al., EndoAIM (2024) [15]	Effective attention-based multi-task transformer, but computationally heavy and hard to execute in resource-limited environments	Need of lightweight transformer structures with less computational load
Siddiqui et al., Deep Ensemble Learning (2025) [10]	Ensemble CNNs improve accuracy but increase inference cost and offer limited transparency	Need for efficient single-model solutions that maintain accuracy while improving explainability

Study, Method (Year)	Limitations Identified	Resulting Research Gap
Masum et al., Ensemble Transformer with XAI (2025) [20]	Ensemble ViT/Swin models provide explainability but are large, complex, and unsuitable for real-time use	Need for lightweight explainable ViTs designed for practical clinical deployment

III. METHODOLOGY

This paper presents a lightweight ViT EndoLite-ViT to classify endoscopic images into four categories: GERD, GERD Normal, Polyp, and Polyp Normal. The section would comprise a combination of data preprocessing, model training, and explainable AI to be sure about precise, efficient, and interpretable predictions. The proposed methodology outlined in Fig. 1 will be followed.

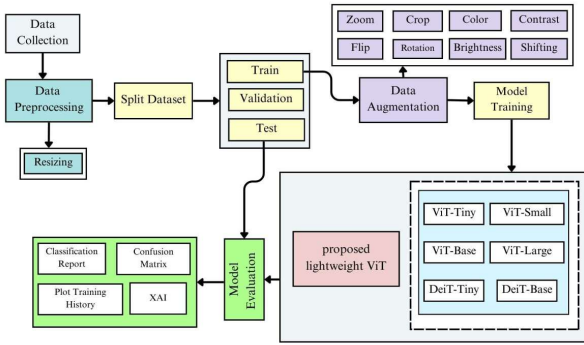


Fig. 1. Proposed Methodology.

A. Data Collection

The study utilized GastroEndoNet (Mendeley Data, February 27, 2025, Version 3) as its data as it has 4,006 original endoscopic images, captured in clinics using medical-grade equipment [23]. The data can be split into four classes, i.e., GERD (974 images), GERD Normal (1,103), Polyp (779) and Polyp Normal (1,150). All the annotations were verified by gastroenterologists and not distorted. The samples of each class are provided in Fig. 2.

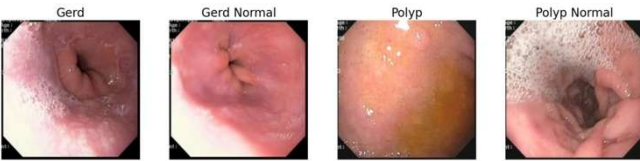


Fig. 2. Visual Image of the Dataset.

B. Data Preprocessing

The GastroEndoNet dataset images had light variations and uneven image quality. To sharpen up the images and highlight pathological characteristics of mucosal texture and polyp shape, a series of preprocessing procedures was used. These were manipulations of brightness, contrast, saturation, and blurring, normalisation of pixel values between 0 and 1, scaling images to 224 x 224 pixels, and noise removal through median filtering. The steps enhanced the quality of the image, lessened noise, and enabled better model generalization of early-stage GERD and polyp detection (Fig. 3).



Fig. 3. Preprocessed Images.

C. Data Augmentation

GastroEndoNet dataset had class imbalance, GERD (974), GERD Normal (1,103), Polyp (779), and Polyp Normal (1,150). To this end, data augmentation was done exclusively to the training set, maximizing underrepresented classes to 2,500 images to the class. The methods of augmentation were random rotations ($\pm 30^\circ$), horizontal/vertical flip, brightness/contrast ($\pm 20\%$), zoom ($0.8 \times 1.2 \times$), translation ($\pm 10\%$), and Gaussian blurring (augmentation to mimic clinical variation). This method balanced the data, better focused the model on pathology features, and increased the overall generalization in the variety of clinical conditions. The general dataset distribution is presented in Table II.

TABLE II. DATASET DISTRIBUTION

Class Name	Original Count	Augmented Training Set
GERD	974	2500
GERD Normal	1103	2500
Polyp	779	2500
Polyp Normal	1150	2500
Total	4006	10000

D. Proposed EndoLite-ViT Model Architecture

The input image of the proposed EndoLite-ViT of size 224x224x3 is split into 8x8 nonexploitation patches by the Patches layer to obtain 784 patches per image with 192 elements each. The patches are learned as a 64-dimensional embedding with the PatchEncoder, then positional embeddings are introduced to store spatial information. These embeddings go through six transformer blocks, each with multi-head self-attention (8 heads) and MLP with GELU activation, residual connections, layer normalization as well as dropout. The four-class output is obtained through the use of global average pooling and then densely layered. Fig. 4 is a visualization of an input image, the 784 patches comprising it, and the patch embeddings, showing the way the model encodes local features to find GERD and polyp.

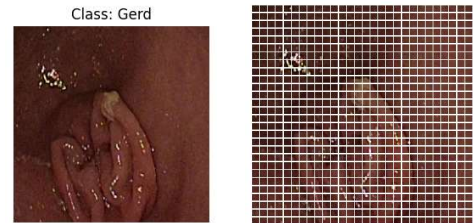


Fig. 4. Patch embeddings of an endoscopic image.

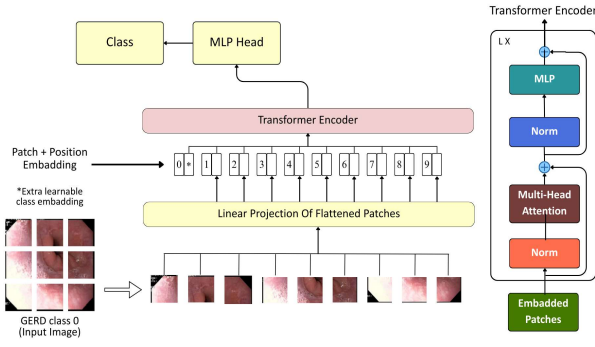


Fig. 5. EndoLite-ViT Model Architecture.

IV. RESULT ANALYSIS DISCUSSION

This part measures the performance of the suggested EndoLite-ViT with the aid of the balanced and augmented GastroEndoNet dataset. The most important metrics will be computed to determine the effectiveness of the model, such as accuracy, precision, recall, and F1-score. Confusion matrices, ROC curves, training/validation accuracy and loss curves are detailed to demonstrate the learning behavior and reliability of the model to classify the image of GERD and colorectal polyp.

A. Performance metrics

Standard classification measures common to machine learning such as Accuracy, Precision, Recall and F1-Score were used to evaluate the performance of each model. The equations of these measures are as follows:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

$$F1\ Score = 2 * \frac{Precision*Recall}{Precision+Recall} \quad (4)$$

Trained and evaluated on the preprocessed, balanced dataset, five ViT-based deep learning models together with the proposed EndoLite-ViT were summarised in Table III in terms of their macro averaged precision, recall, F1-score and accuracy.

B. Performance Evaluation

Five ViT-based deep learning models, along with the proposed EndoLite-ViT, were trained and tested on the preprocessed, balanced dataset, and their macro-averaged precision, recall, F1-score, and accuracy are summarized in Table III.

TABLE III. PERFORMANCE COMPARISON OF ViT-BASED MODELS

Model	Size (Params)	Precision	Recall	F1-Score	Accuracy
ViT-Tiny	20.93M	0.83	0.83	0.83	0.83
ViT-Small	82.36M	0.93	0.93	0.93	0.93
ViT-Large	1156.25M	0.97	0.97	0.97	0.97
DeiT-Tiny	20.93M	0.87	0.87	0.87	0.87
DeiT-Base	326.73M	0.95	0.95	0.95	0.95
EndoLite-ViT (Proposed)	0.18M	0.94	0.95	0.95	0.95

The suggested EndoLite-ViT attained accuracy of 0.95, precision of 0.94 and recall and F1-score of 0.95 with only 0.18M parameters. According to the summary presented in Table III, ViT-Large achieved a marginally better accuracy

(0.97) but with over 1156M parameters thus being inapplicable in resource-limited clinical settings. Similarly, ViT-Small (82.36M; 0.93 accuracy) and DeiT-Base (326.73M; 0.95 accuracy) have significantly more computing needs and do not have an apparent performance benefit over the suggested model. The limited number of parameters of EndoLite-ViT can also be explained by the fact that it has a shallow transformer and a compact patch embedding, combined with the lightweight self-attention block, which minimizes the redundancy in the computations but does not remove the required feature representations. All of these architectural features, along with task-specific training on GastroEndoNet data, allow EndoLite-ViT to have a desirable strike between predictive and computational efficiency, positively contributing to its applicability in real-time and low-resource clinical environments.

TABLE IV. FOLD-WISE PERFORMANCE OF ENDOLite-ViT MODEL

Fold	Precision	Recall	F1-Score	Accuracy
Fold 1	0.9397	0.9354	0.9356	0.9354
Fold 2	0.9444	0.9437	0.9437	0.9437
Fold 3	0.9543	0.9535	0.9533	0.9535
Fold 4	0.9589	0.9597	0.9598	0.9594
Fold 5	0.9697	0.9690	0.9694	0.9698
Average	0.9538	0.9528	0.9529	0.9511

To validate the proposed EndoLite-ViT, now, a 5-fold cross-validation was now considered to make sure that the model is robust and stable. Table IV shows the fold-wise results of performance using precision, recall, F1-score, and accuracy. It can be seen that the fold-wise analysis indicates that EndoLite-ViT has been able to perform well in all its folds with an average precision (0.9538), recall (0.9528), F1-score (0.9529), and accuracy (0.9511). The low difference between folds shows that it has a high degree of generalization and stability of the model, and can be used in practice.

Further, for evaluating the training behavior and stability of the suggested EndoLite-ViT, on the cross-validation, the training and validation accuracy and loss curves of all five folds were plotted in Fig. 6 and Fig. 7. Fig. 6 shows the training and validation accuracy of all the folds. The curves exhibit a smooth convergence and the minimum fluctuation, and the validation accuracy is close to training accuracy without any overfitting. The vast majority of folds gradually become more and more effective and stabilize at around 90 percent, indicating the stability of EndoLite-ViT regarding various subsets of the data.

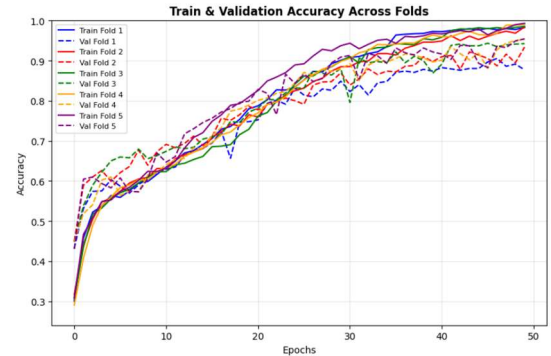


Fig. 6. Training and Validation Accuracy across all folds of EndoLite-ViT.

The training loss curve and the validation loss curve are shown in Fig. 7. The analysis of all folds demonstrates a

gradual reduction of training loss, and validation loss is decreasing steadily without any sharp spikes, which emphasizes the ability of the model to resist overfitting. Both training and validation loss decreased towards the end, which demonstrates that EndoLite-ViT is a good model capable of cross-fold generalization.

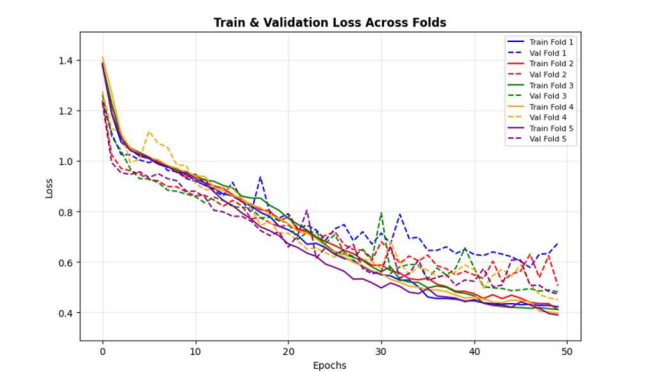


Fig. 7. Training and Validation loss across all folds of EndoLite-ViT.

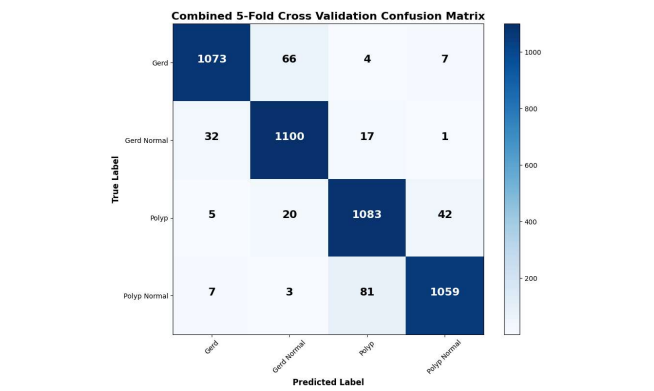


Fig. 8. Ensemble confusion matrix of all folds of EndoLite-ViT.

After the curves of accuracy and loss, a composite confusion matrix of all folds was created to present an integrated picture of the classification results (Fig. 8). The findings indicate that the model was highly recognized, with each class efficiently finding 1073 GERD, 1100 GERD Normal, 1083 Polyp, and 1059 Polyp Normal samples. There were a few cases of misclassifications, including 81 Polyp Normal cases being classified as Polyp, showing minor overlaps, although overall reflecting strong discriminative ability.

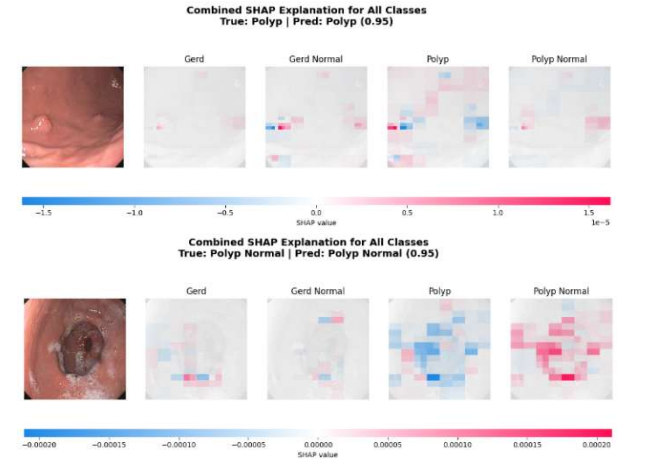


Fig. 9. SHAP analysis across all classes of EndoLite-ViT.

According to the analysis, SHAP analysis was used to visualize the impact of various image parts to the model predictions (Fig. 9). The emphasized areas were quite similar to the characteristics of the classes, which provided an idea about the decision-making process of the model. SHAP gives a measure of feature importance on a global scale, whereas LIME gives local explanations on how individual predictions in the model were made by identifying regions that had the greatest influence on the output. The LIME analysis revealed that the model always gave its attention to the pathological areas- diseased mucosa and polyp forms- and suppressed unnecessary background data. This finding suggests that EndoLite-ViT is based on clinically significant attributes and not spurious trends therefore, increasing the predictability and accuracy of its forecasts (Fig. 10).

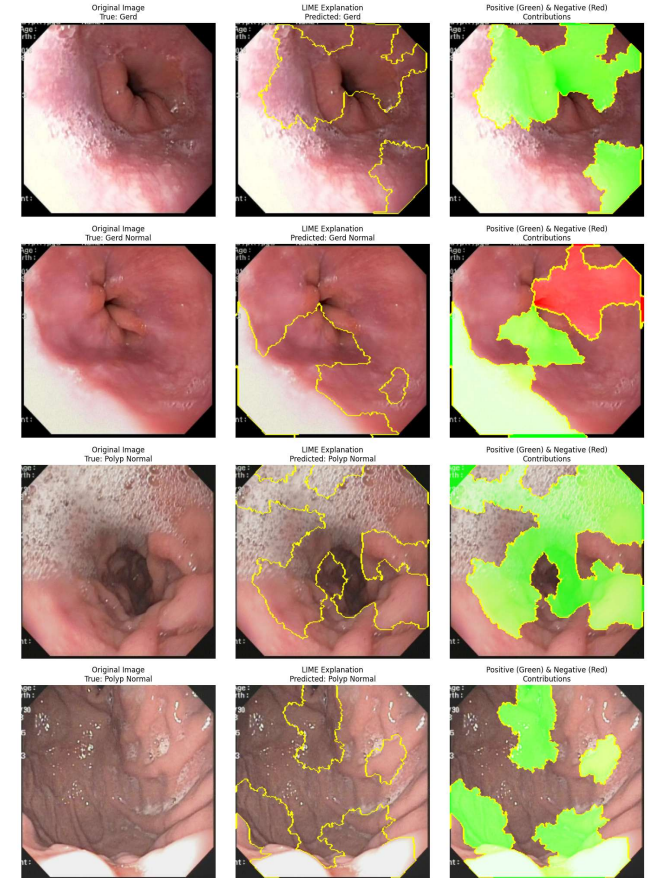


Fig. 10. LIME analysis across all classes of EndoLite-ViT.

V. DISCUSSION

The proposed EndoLite-ViT model achieved good performance mainly because of its lightweight design, which is effective at extracting key features by patch-based attention without over-fitting and can achieve high accuracy with much fewer parameters compared to more traditional large ViTs. Its architecture strikes a balance between complexity and generalization, enabling good learning of the filtered balanced data. The combination of SHAP and LIME presented critical interpretability, which discloses that the model concentrates on clinically significant areas like polyps and lesions as opposed to insignificant background characteristics, which builds confidence and interpretability in medical judgments. Although these are encouraging outcomes, there are some shortcomings. Although balanced, the dataset comprises fairly similar types of images, which may overvalue the performance measures. As well, external validation using more diverse or complex data is not present. Furthermore, practical implementation might demand strength testing in diverse imaging circumstances, and additional evaluation is necessary to determine generalizability to distinct patient groups and endoscopy equipment.

VI. CONCLUSION AND FUTURE WORK

This paper introduces EndoLite-ViT, a light vision transformer model to classify images of endoscopy correctly. The model has been trained on a balanced dataset, and its performance was high, like in much larger ViT architectures, with a fraction of the parameters. K-fold cross-validation ensured consistency in the results among cross-folds, and the combined confusion matrix analysis revealed strong classification in all the classes. The techniques of interpretability, discussed SHAP and LIME, proved that the model concentrates on disease-related areas, which are polyps and lesions, providing reliable predictions. In general, EndoLite-ViT is an efficient, accurate, and explainable tool that can be used in practice when computational resources are scarce.

Future directions involve testing the model using bigger and more heterogeneous data in order to guarantee validity and generalizability. More lightweight architectures can be investigated, and more explainability approaches could be added. Clinical feedback on the domain, as well as the assessment of real-time deployment conditions, could also be helpful to increase the applicability of the model to real-world endoscopic practice.

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